

SimpleTracking Instructions

SimpleTracking Version 2.0

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This is the **first step** of chemotaxis analysis in I2Tumbles. SimpleTracking.m converts a series of .tif images into a .mat file and an excel file that contains X and Y coordinates of the bacterial movements. If you can't find the right setting from SimpleTracking.m, we recommend using [TrackingGUI](#) and use the resulting .mat file for the tumble analysis (CellTrackAnalysis.m).

Step 0: Adjusting Videos to work in SimpleTracking

Before running SimpleTracking, bacterial movement videos should be converted to .tif image files. See Kailie's tutorial [here](#) to learn how to convert videos to image files and parameter settings.

Step 1: SimpleTracking Parameters

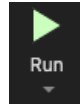
1. Open MATLAB and open SimpleTracking.m
2. Adjust Main parameters. The most important parameters are numFrames and secondVideo. Make sure those two parameters are correct.

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
%% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
%% MAIN PARAMETERS  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
bpfiltsize=      20;           % bpfiltsize      [A]  
hthreshopt=       2;           % Threshold Option [B] 2. Std. Thresh.  
defaultthresh2=   4.0;         % thr (>1)       [C]  
  
stdevMinThreshold= 2;           % Cull: Std. Min. Threshold  
numFrames=        200;         % Cull: # of Video Image Frames  
secondVideo=       10;         % Cull: Length of Video (sec)  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

3. Adjust other parameters if necessary. All the parameters are pre-set to the setting that is frequently used in the lab. See Kailie's tutorial or TrackingGUI [manual](#) if you are unsure which parameters you should use.

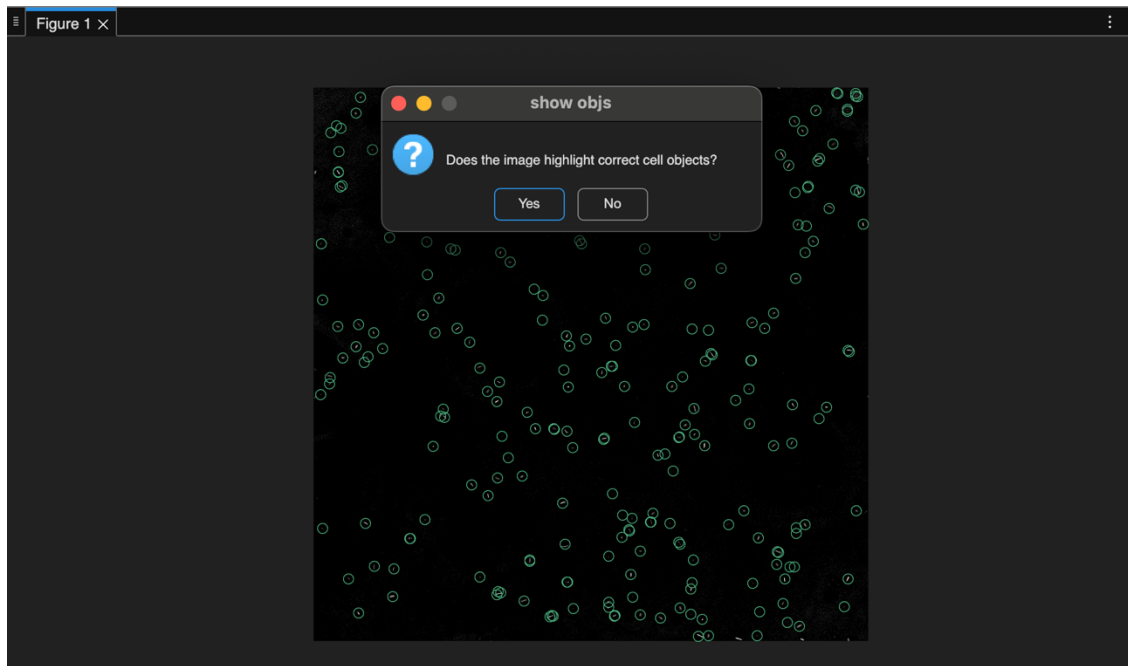
Step 2: Select Images

1. Click Run
2. The script will prompt you to select the image files. Browse to the folder with the images
3. Select the first image (format should be .tif)
4. Select the last image



Step 3: Confirm Objects

1. The first image of the video will pop up, with green circles on each bacterium object to show that the software will track them.
2. Click “Yes” if the objects look correct. Otherwise, click “No” and the software will stop. Adjust parameters and re-run if incorrect.



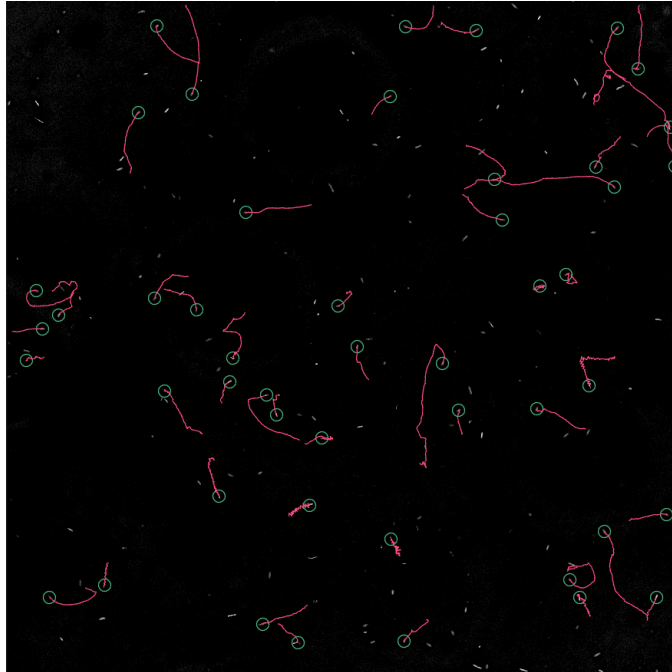
Step 4: Track, Link and Cull Objects

1. After confirming, the software will proceed with tracking, Linking and Culling all objects. You can see the progress with the progress bar.
2. If parameters aren't changed, the software will cull twice (Stdev, TrackLength)
3. You'll be able to see the number of tracks after culling on the Command Window. Copy the number if needed. In the screenshot, 6036 is # tracks before the cull and 284 is after.

```
>> SimpleTracking
Found 6036 unique tracks.
More than 10 tracks found; not displaying details.
Number of Tracks After Cull:
284
```

Step 5: Save Objects

1. Next, another popup will show with the object tracks after the cull with red lines. Confirm that the tracks look correct.



2. SimpleTracking saves the output and automatically runs Kailie's additional scripts and `migration_v2.m` afterwards.
3. There will be two outputs:
 - a. `SimpleTrackingoutput.mat`
 - b. `Tracked_positions.xlsx`

Context and References

SimpleTracking is a MATLAB script written for Baylink Lab (www.baylink-lab.com) to simplify bacteria tracking from videos and provide bacteria movement metrics, utilizing TrackingGUI scripts consolidated with other scripts written by Baylink lab members. Please refer to the links below for further information:

- TrackingGUI GitHub: https://github.com/rplab/TrackingGUI_and_Localization_Public
- ImageJ and TrackingGUI instructions by Kailie Franco: https://github.com/clara-h-shin/SimpleTracking/blob/main/swim_tracking_tutorial_by_kailie.docx